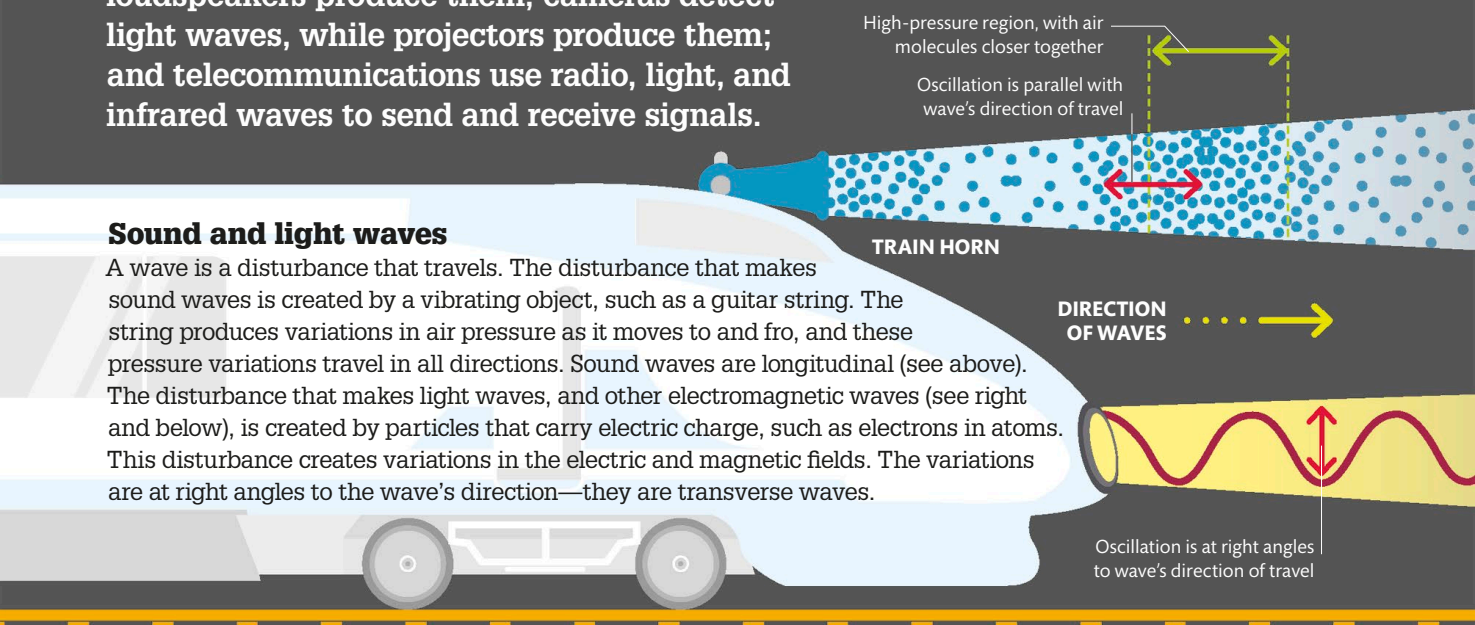
Waves

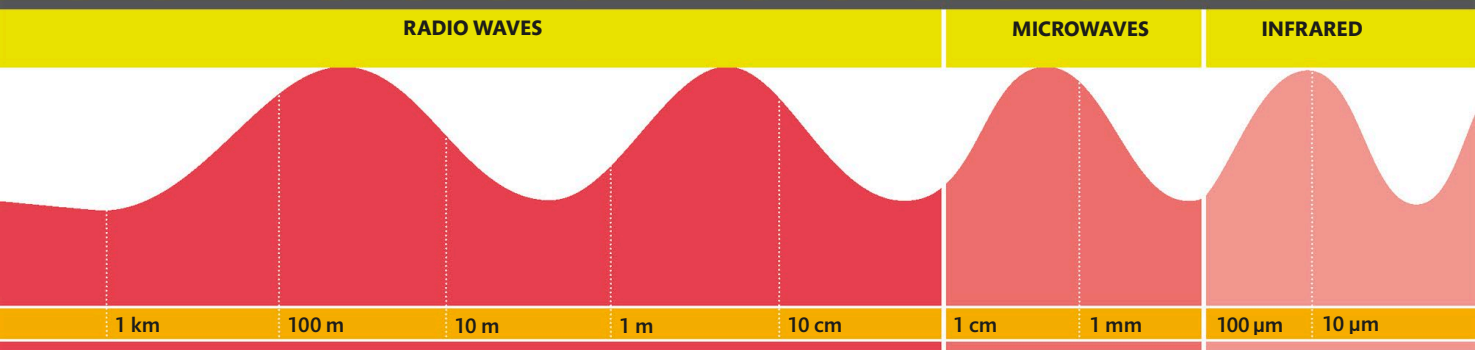
Many technologies work with waves: microphones detect sound waves, while loudspeakers produce them; cameras detect light waves, while projectors produce them; and telecommunications use radio, light, and infrared waves to send and receive signals.

Longitudinal wave
Sound waves are longitudinal. This is because the variation of air pressure is backward and forward, going in the same direction as the wave is traveling.



Sound and light waves

A wave is a disturbance that travels. The disturbance that makes sound waves is created by a vibrating object, such as a guitar string. The string produces variations in air pressure as it moves to and fro, and these pressure variations travel in all directions. Sound waves are longitudinal (see above). The disturbance that makes light waves, and other electromagnetic waves (see right and below), is created by particles that carry electric charge, such as electrons in atoms. This disturbance creates variations in the electric and magnetic fields. The variations are at right angles to the wave's direction—they are transverse waves.



The electromagnetic spectrum

Light is electromagnetic radiation—waves created by disturbances in electric and magnetic fields. Our eyes are sensitive to a range of light from lower-frequency red light to higher-frequency blue light. But there are other kinds of electromagnetic radiation beyond the visible spectrum: radio waves, microwaves, and infrared radiation, with lower frequencies than visible light, and ultraviolet radiation, X-rays, and gamma rays, with higher frequencies.



Radio telescope
A dish antenna can be used to detect radio waves emitted by distant stars.



Microwave oven
Food heats up when high-energy microwaves excite the water molecules inside.



Remote control
A remote control uses pulses of infrared radiation to transmit digital control codes.